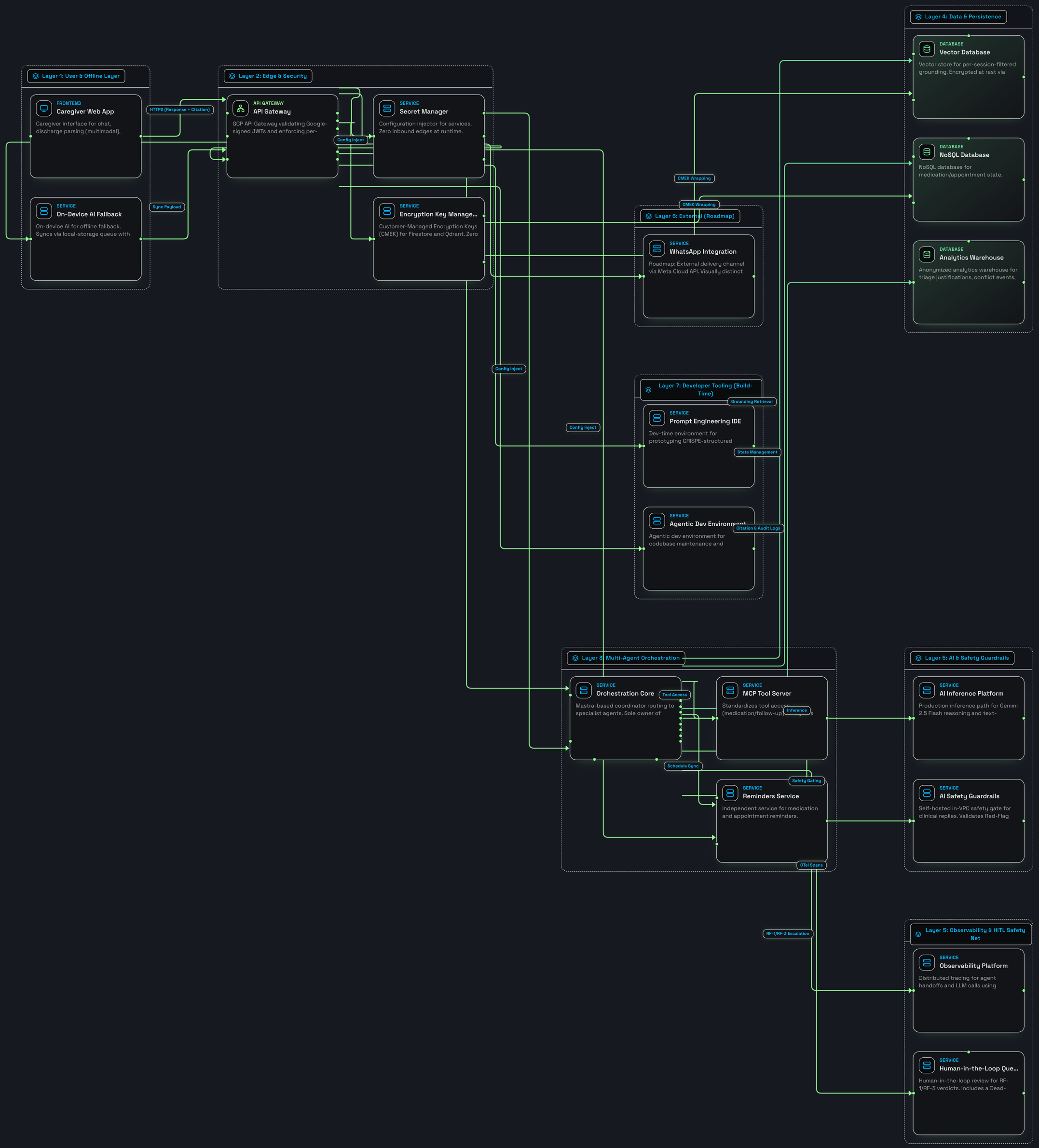
Layer 1: User & Offline Layer
FRONTEND
Caregiver Web App
Caregiver interface for chat, discharge parsing (multimodal),
SERVICE
On-Device AI Fallback
On-device AI for offline fallback. Syncs via local-storage queue with
HTTPS (Response + Citation)
Sync Payload
Layer 2: Edge & Security
API GATEWAY
API Gateway
GCP API Gateway validating Google-signed JWTs and enforcing per-
SERVICE
Secret Manager
Configuration injector for services. Zero inbound edges at runtime.
SERVICE
Encryption Key Manage...
Customer-Managed Encryption Keys (CMEK) for Firestore and Qdrant. Zero
Config Inject
Config Inject
Config Inject
Config Inject
Layer 4: Data & Persistence
DATABASE
Vector Database
Vector store for per-session-filtered grounding. Encrypted at rest via
DATABASE
NoSQL Database
NoSQL database for medication/appointment state.
DATABASE
Analytics Warehouse
Anonymized analytics warehouse for triage justifications, conflict events,
CMEK Wrapping
CMEK Wrapping
Layer 6: External (Roadmap)
SERVICE
WhatsApp Integration
Roadmap: External delivery channel via Meta Cloud API. Visually distinct
Layer 7: Developer Tooling (Build-Time)
Grounding Retrieval
SERVICE
Prompt Engineering IDE
Dev-time environment for prototyping CRISPE-structured
State Management
SERVICE
Agentic Dev Environment
Agentic dev environment for codebase maintenance and
Citation & Audit Logs
Layer 3: Multi-Agent Orchestration
SERVICE
Orchestration Core
Mastra-based coordinator routing to specialist agents. Sole owner of
Tool Access
SERVICE
MCP Tool Server
Standardizes tool access (medication/follow-up) ...gs
Inference
Schedule Sync
Safety Gating
SERVICE
Reminders Service
Independent service for medication and appointment reminders.
OTel Spans
Layer 5: AI & Safety Guardrails
SERVICE
AI Inference Platform
Production inference path for Gemini 2.5 Flash reasoning and text-
SERVICE
AI Safety Guardrails
Self-hosted in-VPC safety gate for clinical replies. Validates Red-Flag
RF-1/RF-3 Escalation
Layer 5: Observability & HITL Safety Net
SERVICE
Observability Platform
Distributed tracing for agent handoffs and LLM calls using
SERVICE
Human-in-the-Loop Que...
Human-in-the-loop review for RF-1/RF-3 verdicts. Includes a Dead-